

# Luke Cullen

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## EDUCATION

### University of Central Florida

Bachelor of Science in Computer Science - Minor in Data Science - 3.72 GPA

Orlando, FL

August 2023 – May 2027

## EXPERIENCE

### Web Developer Intern

October 2025 – Present

UCF Techrangers

Orlando, FL

- Recently accepted an offer for the **UCF Techrangers**, an organization which develops **educational software** that impacts **10,000+ students annually**.
- Expected to **onboard and maintain** pre-existing educational solutions alongside a team of **27 developers and fellow interns**.
- Taking initiative to **brainstorm, plan and implement** new tools for University students and faculty.
- Developing **methods** and **marketing pitches** to **increase adoption** of our tools across the University.

### Hackathon Organizer

July 2025 – Present

Knight Hacks

Orlando, FL

- Working with our directing team to utilize over **\$60,000** worth of event funding.
- Constructing an environment that provides hundreds of student hackers deadline-limited programming experience.
- Assisting in logistics for a target of **1000+** people attending the event.
- Planning engaging events to combat development burnout and make Knight Hacks truly memorable.

### Lead Developer, Vice President

May 2025 – Present

DenTraceX

Orlando, FL

- **Co-founded a non-profit organization** dedicated to collecting dental records of missing persons, in order to cross-reference them with unidentified remains.
- Authored a **full-stack React application** from scratch, using **ExpressJS**, **PostgreSQL**, as well as our own self-hosted **Minio S3 bucket** storage solution.
- Constructed a **novel** code-based representation for jaw/tooth records, allowing easy searching/filtering to identify **real-world** human remains, intended for use by medical examiners and independent investigators.
- Working with a team of diverse disciplines to process and handle **sensitive medical information** and **maintain legal compliance**.

## PROJECTS

### VL-ADK | Google ADK, NVIDIA Jetson Orin, NVIDIA NemoTron, YOLO-E, React, FastAPI

September 2025

- Created at ShellHacks 2025, won **1st place for NVIDIA**, **2nd place for Microsoft**, and **3rd place for Waymo** and their respective sponsor challenges.
- Mimicks the functionality of a **generalized VLA robotics model** using a fleet of **Google ADK** agents, powered by **NVIDIA NemoTron**.
- Allows **dynamic prompting** of a cost-effective **NVIDIA Jetson Orin**-powered robot, achieving **intelligent movement and spatial reasoning**, even with extremely limited sensing capabilities.
- With the assistance of a **YOLO-E** model for object detection, accomplished **structured paths** and **object search/navigation** behaviors, completely operated by **autonomous AI agents**.

### Voiceboard | React, TypeScript, Gemini API, Mermaid

September 2024

- Created at ShellHacks 2024, won **1st place for Google's Hack for Social Good Challenge**.
- Aims to make whiteboarding accessible to all using **speech recognition** and Google's **Gemini AI**.
- Implements prompt engineering through the **Gemini API** to create **Mermaid diagram code**.
- Developed a **context aware system** to allow users to continually iterate on diagrams.

### Lootcode | React, Clerk, TypeScript, MySQL, Zx Shell, Docker, tRPC

February 2024 – May 2024

- Created for **Knight Hacks Project Launch 2024**, won **1st place**
- Developed a **gamified study application** for Data Structures and Algorithms.
- Live-generates **Docker containers** to self-host code grading instances, maintaining **server security**
- **7000+** unique visits, featured in **Linux Magazine's** OSS section

## TECHNICAL SKILLS

**Languages:** Java, Python, C/C++/C#, Lua, SQL, TypeScript, HTML/CSS

**Libraries/Frameworks:** React, ExpressJS, FastAPI, S3, Prisma, Clerk, JWT, tRPC, MySQL, PostgreSQL, Node.js, Tailwind CSS, Websockets, OpenCV, MediaPipe, pandas, YOLO-E, Google Gemini, NVIDIA NemoTron, Google ADK

**Developer Tools:** Linux, Coolify, Git, Docker, VS Code, Eclipse, Cursor AI